FACTS - Facts Authenticity and Credibility Tracking System: EBSI Simulator

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1. Project Overview

1.1 Project Description

This part of the project aims to simulate the EBSI API's in order to make it possible to build and test applications of EBSI.

1.2 Technology Stack

Component	Technology
Backend Framework	FastAPI (Python)
Database	PostgreSQL with SQLModel/SQLAlchemy

1.3 EBSI Blockchain Simulation

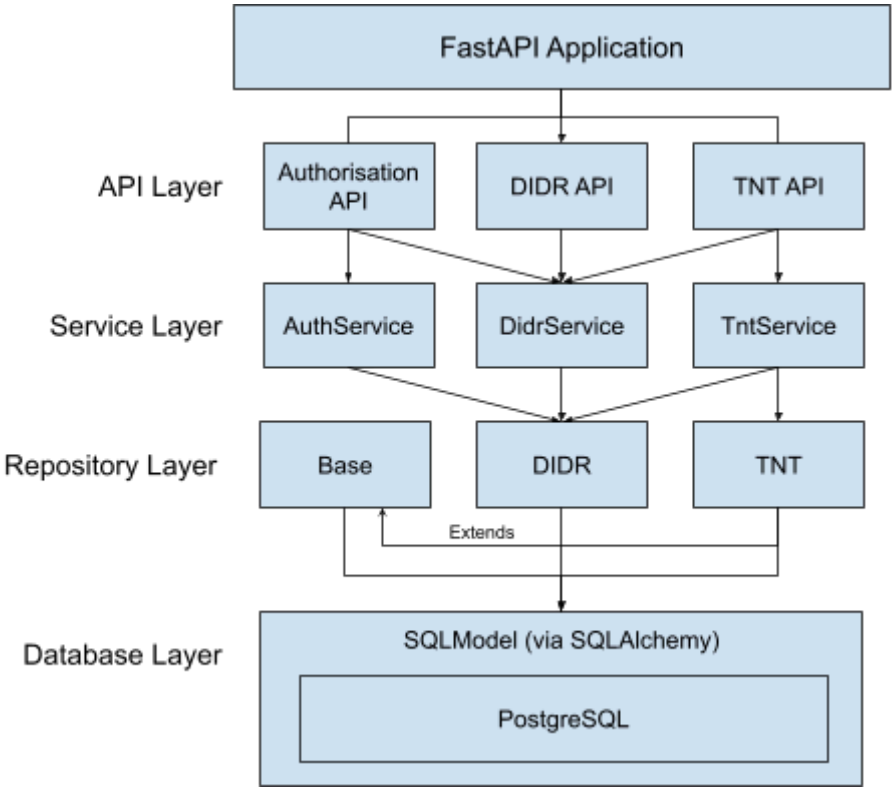
The EBSI Blockchain is simulated using FastAPI and a PostgreSQL database, including all the CRUD actions. The simulation includes:

- Authorisation API (v4) - OAuth2 and OIDC implementation
- Track'n'Trace API (TNT) - Document and event tracking
- DID Registry API (v5) - Decentralized Identifier management
- Wallet Mock - Verifiable Presentation creation and ETH Transaction signature
- Issuer Mock - Verifiable Credential issuance

2. Architecture

The simulator is built following the "**Service - Repository**" design pattern.

2.1 System Architecture Diagram (excluding mock APIs)



2.2 Layer Descriptions

API Layer (ebsi_sim/api/)

- Handles HTTP requests and responses
- Implements RESTful endpoints for each service
- Handles authentication and authorization
- Input validation using Pydantic schemas

Service Layer (ebsi_sim/services/)

- Contains business logic
- Implements use cases and business rules
- Coordinates between API and Repository layers
- Handles transaction management

Repository Layer (ebsi_sim/repositories/)

- Implements Data Access Object (DAO) pattern
- Provides generic CRUD operations
- Abstracts database operations
- Returns domain models to service layer

Model Layer (ebsi_sim/models/)

- Defines database table schemas
- Uses SQLModel for ORM functionality
- Defines relationships between entities
- Includes database constraints

3. API Endpoints

3.1 Authorisation API (/authorisation)

Method	Endpoint	Description
GET	/.well-known/openid-configuration	OpenID Provider Metadata
GET	/jwks	OpenID Provider's public keys
GET	/presentation-definitions	Presentation definition requirements
POST	/token	Token endpoint for OAuth2 flow

3.2 DID Registry API (/did-registry)

Method	Endpoint	Description
GET	/abi	Get smart contract ABI
GET	/identifiers	List identifiers
GET	/identifiers/{did}	Get a DID document
POST	/json-rpc	JSON-RPC endpoint

3.3 Track and Trace API (/track-and-trace)

Method	Endpoint	Description
GET	/abi	Get smart contract ABI
GET	/documents	List documents
GET	/documents/{document_id}	Get specific document
GET	/documents/{document_id}/events	List document events
GET	/documents/{document_id}/events/{event_id}	Get a document event
GET	/documents/{document_id}/accesses	List document accesses
GET	/accesses	List accesses
HEAD	/accesses	Check if DID has creator access
POST	/json-rpc	JSON-RPC endpoint

3.4 Mock APIs

Wallet Mock API (/wallet-mock)

Method	Endpoint	Description
GET	/create_vp	Create Verifiable Presentation from credential
GET	/sign_transaction	Sign Ethereum transaction

Issuer Mock API (/issuer-mock)

Method	Endpoint	Description
GET	/request_vc	Request Verifiable Credential

4. Service Layer

4.1 AuthService (ebsi_sim/services/authorisation.py)

- generate_access_token() - Generate JWT access token
- generate_id_token() - Generate JWT ID token
- load_presentation() - Load presentation definition by scope
- decode_and_check_signature() - Decode and verify token signature
- find_credential() - Find and validate credential in VP
- extract_and_validate_credentials() - Extract credentials from VP
- get_verifiable_presentation() - Decode and validate VP
- check_scope_constraints() - Validate scope constraints
- create_token() - Create access and ID tokens from VP

4.2 DidrService (ebsi_sim/services/didr.py)

- get_did_document() - Retrieve DID document by DID
- count_did_documents() - Count DID documents with optional controller filter
- list_did_documents() - List DID documents with pagination and controller filter
- get_verification_method() - Get verification method by ID
- list_verification_methods() - List verification methods with pagination
- insert_did_document() - JSON-RPC: Create new DID with verification methods and relationships
- update_base_document() - JSON-RPC: Update DID base document
- add_controller() - JSON-RPC: Add DID controller
- revoke_controller() - JSON-RPC: Revoke DID controller
- add_verification_method() - JSON-RPC: Add verification method to Identifier
- add_verification_relationship() - JSON-RPC: Add verification relationship (authentication, assertionMethod, etc.)
- revoke_verification_method() - JSON-RPC: Revoke verification method (set expiration)
- expire_verification_method() - JSON-RPC: Set expiration date for verification method
- get_abi() - Get DID Registry contract ABI
- handle_rpc() - Handle JSON-RPC requests with scope and access validation

JSON-RPC Scope Requirements:

Method	**Required Scopes**
insertDidDocument	didr_invite, didr_write
updateBaseDocument	didr_write
addService	didr_write
revokeService	didr_write
addController	didr_write
revokeController	didr_write
addVerificationMethod	didr_write
addVerificationRelationship	didr_write
revokeVerificationMethod	didr_write
expireVerificationMethod	didr_write
rollVerificationMethod	didr_write
sendSignedTransaction	didr_invite, didr_write

4.3 TntService (ebsi_sim/services/tnt.py)

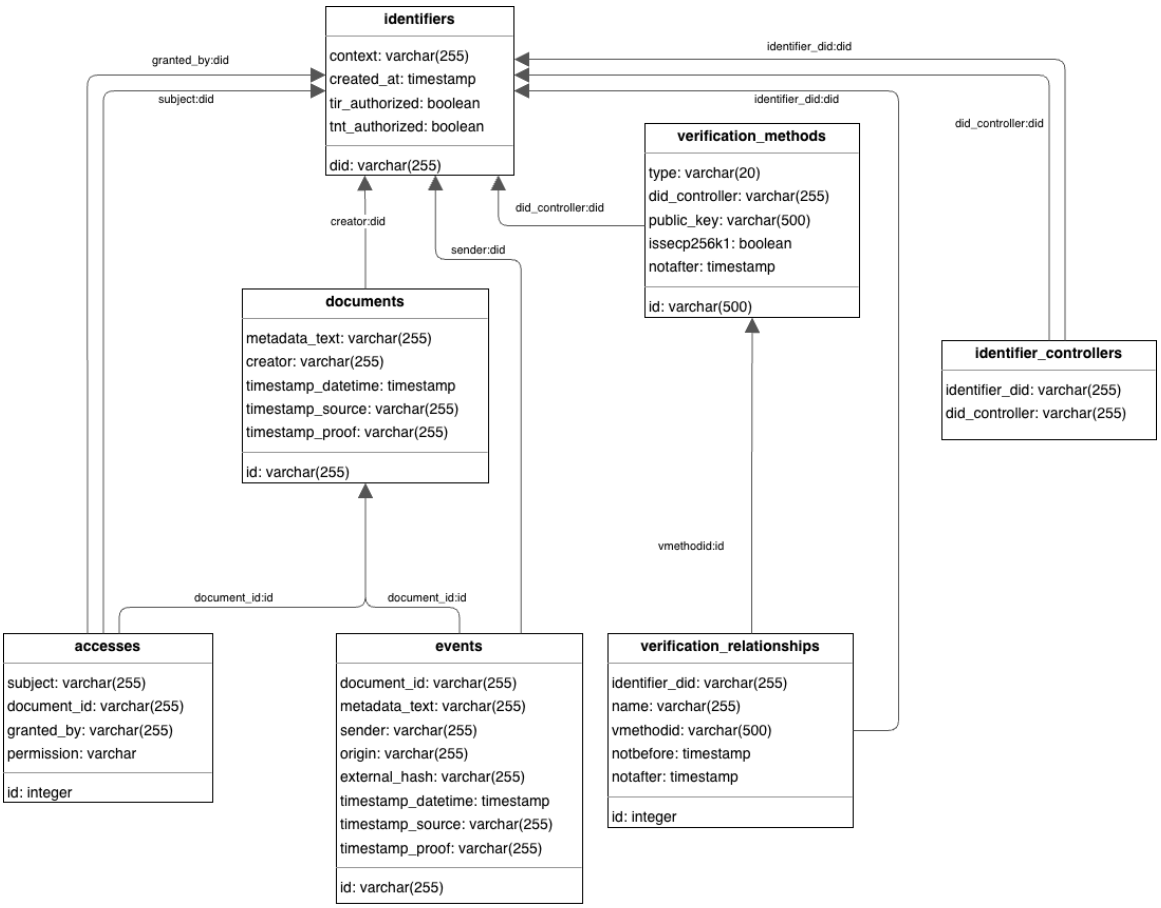
- get_document() - Retrieve document by hash
- count_documents() - Count documents with filters

- list_documents() - List documents with pagination
- count_accesses() - Count accesses with filters
- list_accesses() - List accesses with pagination
- count_events() - Count events with filters
- list_events() - List events with pagination
- authorise_did() - JSON-RPC: Authorize/unauthorize DID for TNT operations
- create_document() - JSON-RPC: Create new document with metadata and timestamps
- remove_document() - JSON-RPC: Remove document (creator only)
- grant_access() - JSON-RPC: Grant write/delegate access to document
- revoke_access() - JSON-RPC: Revoke document access
- write_event() - JSON-RPC: Write event to document with access control
- get_abi() - Get TNT contract ABI
- handle_rpc() - Handle JSON-RPC requests with scope and method constraint validation

JSON-RPC Scope Requirements:

Method	**Required Scopes**
authoriseDid	tnt_authorise
createDocument	tnt_create
removeDocument	tnt_write
grantAccess	tnt_write
revokeAccess	tnt_write
writeEvent	tnt_write
sendSignedTransaction	tnt_authorise, tnt_create, tnt_write

5. Database structure



5.1 TNT Models

- Document - TNT Document
- Access - Document Access
- Event - Document Event

5.2 DIDR Models

- Identifier - Decentralized Identifier entity
- VerificationMethod - Verification method for cryptographic operations
- VerificationRelationship - Relationships between DIDs and verification methods
- IdentifierController - Controller relationships between DIDs

6. Exception Handling

6.1 Core Exceptions (ebsi_sim/core/exceptions.py)

Exception	Description
EBSIError	Base exception for all EBSI errors (Status Code: 500)
EBSIDatabaseError	Database-related errors (Status Code: 500)
EBSIDuplicateError	Duplicate entry errors (Status Code: 409)
EBSIAuthError	Authentication errors (Status Code: 401)
EBSIRequestError	Request processing errors (Status Code: 400)

EBSINotFoundError	Resource not found errors (Status Code: 404)

6.2 Service-Specific Exceptions

- ('AuthServiceError', 'Base error for Auth service operations (Status Code: 500)')
- ('AuthServiceAuthError', 'Authentication error in Auth service (Status Code: 401)')
- ('AuthServiceNotFoundError', 'Resource not found in Auth service (Status Code: 404)')
- ('AuthServiceRequestError', 'Request error in Auth service (Status Code: 400)')
- ('DidrServiceError', 'Base error for DID Registry service (Status Code: 500)')
- ('DidrServiceDuplicateError', 'Duplicate entry in DID Registry service (Status Code: 409)')
- ('DidrServiceAuthError', 'Authentication error in DID Registry service (Status Code: 401)')
- ('DidrServiceNotFoundError', 'Resource not found in DID Registry service (Status Code: 404)')
- ('DidrServiceRequestError', 'Request error in DID Registry service (Status Code: 400)')
- ('TntServiceError', 'Base error for TNT service (Status Code: 500)')
- ('TntServiceDuplicateError', 'Duplicate entry in TNT service (Status Code: 409)')
- ('TntServiceAuthError', 'Authentication error in TNT service (Status Code: 401)')
- ('TntServiceNotFoundError', 'Resource not found in TNT service (Status Code: 404)')
- ('TntServiceRequestError', 'Request error in TNT service (Status Code: 400)')

9. Testing

9.1 Test Files

Test File	Description
test_authorisation.py	Tests for Authorisation API endpoints
test_didr.py	Tests for DID Registry API endpoints
test_tnt.py	Tests for Track'n'Trace API endpoints

9.2 Test Coverage

The test suite covers:

- API endpoint response status codes
- OIDC configuration discovery
- Presentation definitions retrieval
- DID document CRUD operations
- Document retrieval and listing
- Event retrieval and listing
- Access control management
- Authentication endpoints

10. Configuration

10.1 Configuration Variables (ebsi_sim/core/config.py)

Setting	Description
app_name	Application name
db_user	PostgreSQL username
db_password	PostgreSQL password
db_host	PostgreSQL host
db_port	PostgreSQL port
db_name	PostgreSQL database name
AUTH_PUBLIC_KEY	Authorization service public key (ES256)

AUTH_PRIVATE_KEY	Authorization service private key (ES256)
ETH_ADDRESS	Ethereum address
ISSUER_AUTHENTICATION_PUBLIC_KEY	Issuer authentication public key (ES256K)
ISSUER_AUTHENTICATION_PRIVATE_KEY	Issuer authentication private key (ES256K)
ISSUER_ASSERTION_PUBLIC_KEY	Issuer assertion public key (ES256)
ISSUER_ASSERTION_PRIVATE_KEY	Issuer assertion private key (ES256)
ISSUER_DID	Issuer DID

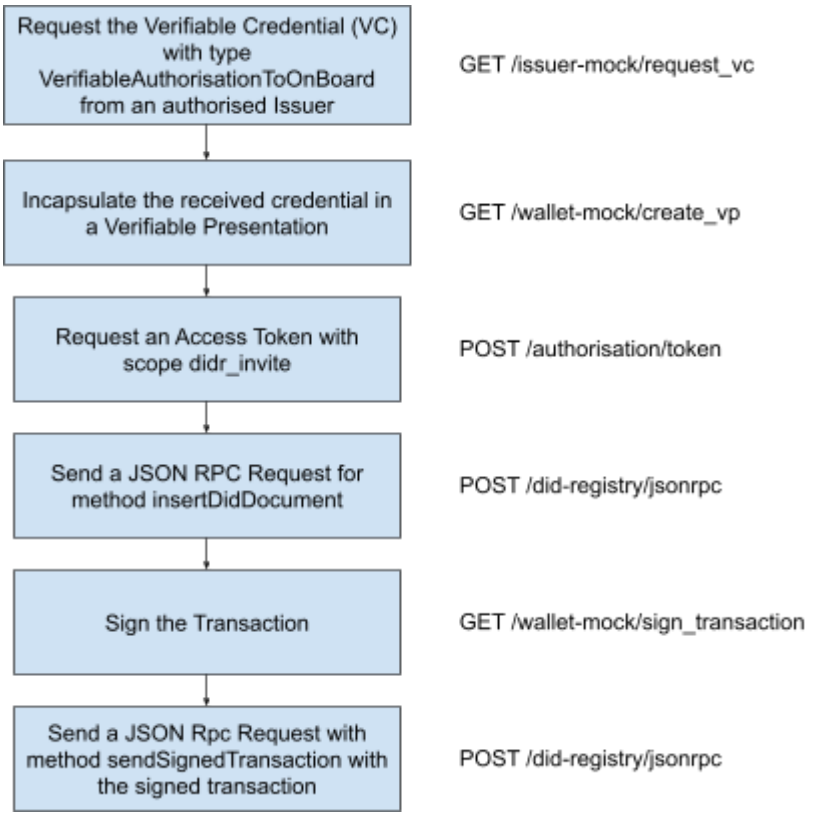
11. Docker Configuration

The project includes Docker configuration for easy deployment and development:

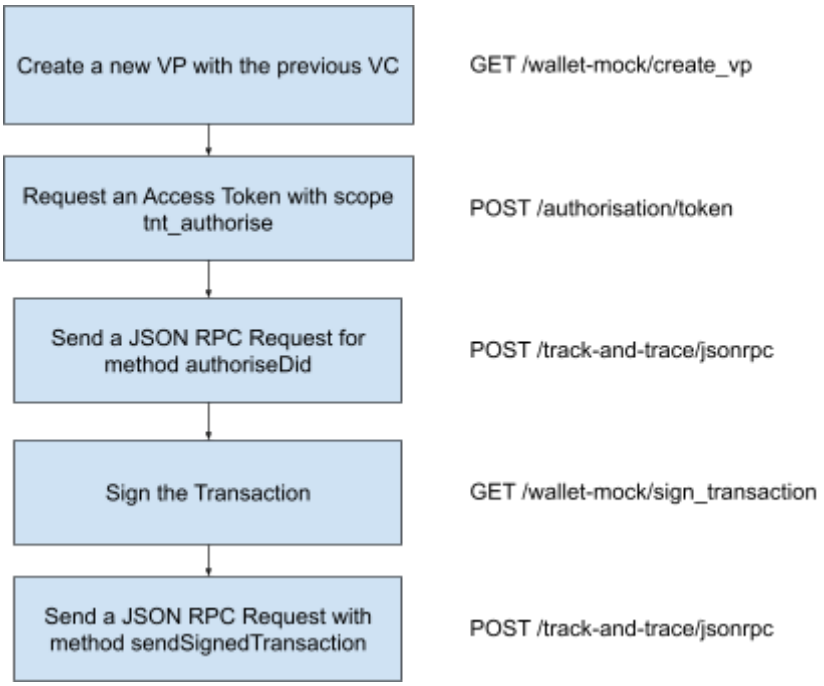
- Dockerfile - Container definition for the EBSI Simulator
- compose.yaml - Docker Compose configuration for multi-container setup

12. Simulator Base Flows

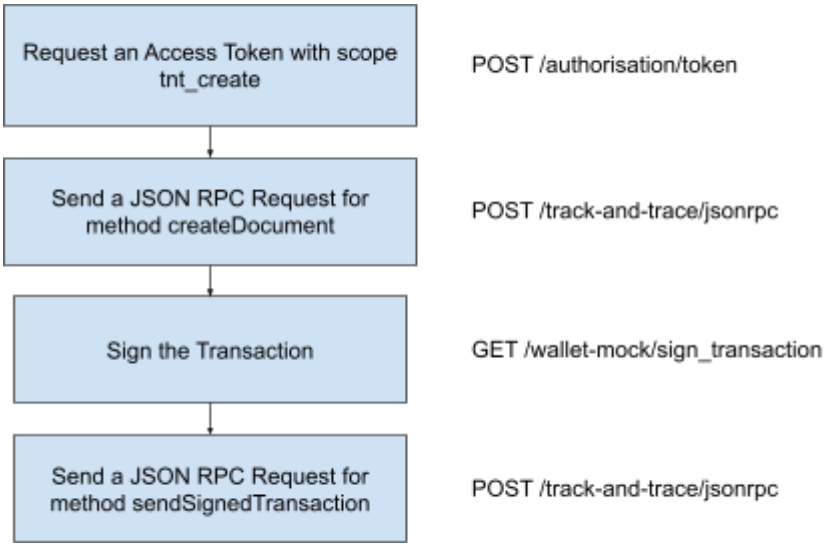
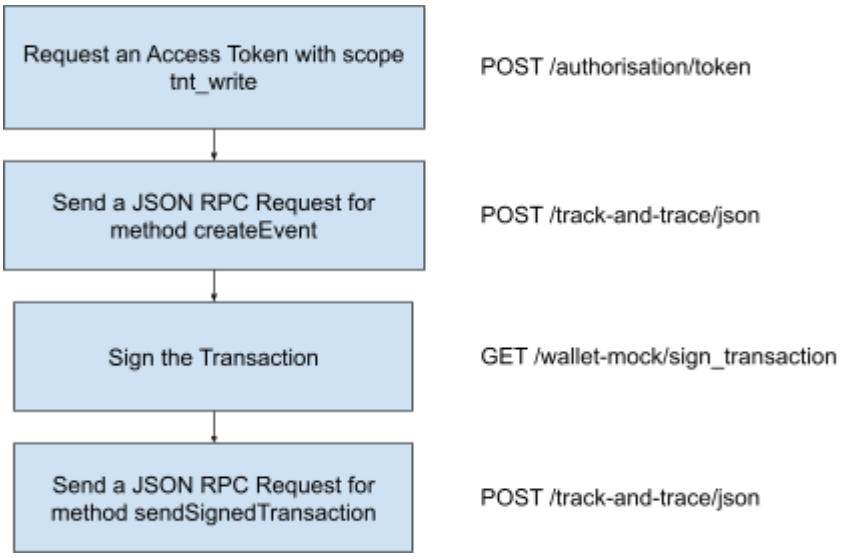
12.1 Create DID Document



12.2 Get authorization to operate on TNT



12.3 Create TNT Document



12.4 Add event to a TNT Document